Double Integrals

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Double integrals over a rectangle**

Set up the double integral of $f(x, y) = xy$ over the rectangle $[0, 2] \times [0, 3]$ (integrate y first).

$\int_0^2 \int_0^3 xy \, dy \, dx$.

2 **[Foundational]** • **Target: Double integrals over a rectangle**

Set up the double integral of $f(x, y) = x^2 + y$ over the rectangle $[1, 3] \times [0, 2]$ (integrate x first).

$\int_0^2 \int_1^3 (x^2 + y) \, dx \, dy$.

3 **[Foundational]** • **Target: Iterated integrals**

Evaluate $\int_0^2 \int_0^3 xy \, dy \, dx$.

Inner: $\int_0^3 xy \, dy = x \cdot \frac{y^2}{2}\big|_0^3 = \frac{9x}{2}$. Outer: $\int_0^2 \frac{9x}{2} \, dx = \frac{9}{4}x^2\big|_0^2 = 9$.

4 **[Foundational]** • **Target: Iterated integrals**

Evaluate $\int_0^1 \int_0^2 (x^2 + y) \, dy \, dx$.

Inner: $\int_0^2 (x^2 + y) \, dy = x^2 y + \frac{y^2}{2}\big|_0^2 = 2x^2 + 2$. Outer: $\int_0^1 (2x^2 + 2) \, dx = \frac{2}{3}x^3 + 2x\big|_0^1 = \frac{8}{3}$.

5 **[Foundational]** • **Target: Iterated integrals**

Evaluate $\int_0^\pi \int_0^1 y \sin x \, dy \, dx$.

Inner: $\int_0^1 y \sin x \, dy = \sin x \cdot \frac{y^2}{2}\big|_0^1 = \frac{\sin x}{2}$. Outer: $\int_0^\pi \frac{\sin x}{2} \, dx = \frac{1}{2}[-\cos x]_0^\pi = \frac{1}{2}(1 - (-1)) = 1$.

6 **[Foundational]** • **Target: Fubini's Theorem: order of integration**

Verify Fubini's Theorem for $f(x, y) = xy$ on $[0, 1] \times [0, 2]$ by evaluating the integral in both orders.

Order $dy \, dx$: inner $= x \cdot \frac{y^2}{2}\big|_0^2 = 2x$, outer $= \int_0^1 2x \, dx = 1$. Order $dx \, dy$: inner $= y \cdot \frac{x^2}{2}\big|_0^1 = \frac{y}{2}$, outer $= \int_0^2 \frac{y}{2} \, dy = 1$. Both orders give 1.

7 **[Foundational]** • **Target: Fubini's Theorem: order of integration**

State the hypothesis of Fubini's Theorem needed for the two iterated orders to be guaranteed equal.

$f(x, y)$ must be continuous (or at least integrable) on the rectangular region $[a, b] \times [c, d]$ — under that condition, the double integral equals either iterated integral, in either order.

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Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Double integrals over general regions

Evaluate $\int_0^1 \int_0^x (x+y)\,dy\,dx$.

Inner: $\int_0^x (x+y)\,dy = xy + \frac{y^2}{2}\big|_0^x = x^2 + \frac{x^2}{2} = \frac{3x^2}{2}$. Outer: $\int_0^1 \frac{3x^2}{2}\,dx = \frac{1}{2}x^3\big|_0^1 = \frac{1}{2}$.

9 [Intermediate] • Target: Double integrals over general regions

Set up (do not evaluate) the double integral of $f(x, y) = xy$ over the triangular region bounded by $y = 0$, $x = 1$, and $y = x$.

$\int_0^1 \int_0^x xy\,dy\,dx$ (for a fixed x between 0 and 1, y ranges from the bottom edge $y = 0$ up to the line $y = x$).

10 [Intermediate] • Target: Double integrals over general regions

Evaluate $\int_0^1 \int_y^1 2\,dx\,dy$ (a region bounded by $x = y$, $x = 1$, $y = 0$).

Inner: $\int_y^1 2\,dx = 2(1-y)$. Outer: $\int_0^1 2(1-y)\,dy = 2\left[y - \frac{y^2}{2}\right]_0^1 = 2\left(1 - \frac{1}{2}\right) = 1$.

11 [Intermediate] • Target: Reversing the order of integration

Reverse the order of integration for $\int_0^1 \int_0^x f(x, y)\,dy\,dx$, describing the new bounds.

The region is $0 \le y \le x \le 1$. Reversed: for a fixed y between 0 and 1, x ranges from y up to 1: $\int_0^1 \int_y^1 f(x, y)\,dx\,dy$.

12 [Intermediate] • Target: Reversing the order of integration

Evaluate $\int_0^1 \int_x^1 e^{y^2}\,dy\,dx$ by first reversing the order of integration (the inner integral has no elementary antiderivative in the original order).

The region is $0 \le x \le y \le 1$. Reversed: $\int_0^1 \int_0^y e^{y^2}\,dx\,dy = \int_0^1 y\,e^{y^2}\,dy$. Substituting $u = y^2$, $du = 2y\,dy$: $= \frac{1}{2}\int_0^1 e^u\,du = \frac{1}{2}(e-1)$.

13 [Intermediate] • Target: Polar coordinates

Convert $\iint_D (x^2 + y^2)\,dA$ to polar coordinates, where D is the disk $x^2 + y^2 \le 4$, and evaluate.

$x^2 + y^2 = r^2$, $dA = r\,dr\,d\theta$. $\int_0^{2\pi} \int_0^2 r^3\,dr\,d\theta$. Inner: $\frac{r^4}{4}\big|_0^2 = 4$. Outer: $\int_0^{2\pi} 4\,d\theta = 8\pi$.

14 [Intermediate] • Target: Polar coordinates

Evaluate $\iint_D xy\,dA$ where D is the quarter disk $x^2 + y^2 \le 1$, $x \ge 0$, $y \ge 0$, using polar coordinates.

$xy = r^2 \sin\theta\cos\theta$, $dA = r\,dr\,d\theta$, $\theta \in [0, \frac{\pi}{2}]$, $r \in [0, 1]$. $\int_0^{\pi/2} \sin\theta\cos\theta\,d\theta \cdot \int_0^1 r^3\,dr = \frac{1}{2} \cdot \frac{1}{4} = \frac{1}{8}$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Polar coordinates • Cross-Lesson: 3.01 Optimization: Critical Points & Lagrange Multipliers

Use polar coordinates to evaluate $\iint_D e^{-(x^2+y^2)}\,dA$ where D is the entire plane $\mathbb{R}^2$ (the classic Gaussian integral setup).

$\int_0^{2\pi}\int_0^{\infty} e^{-r^2} r\,dr\,d\theta$. Inner: let $u = r^2$, $du = 2r\,dr$: $\int_0^{\infty} e^{-r^2} r\,dr = \frac{1}{2}\int_0^{\infty} e^{-u}\,du = \frac{1}{2}$. Outer: $\int_0^{2\pi} \frac{1}{2}\,d\theta = \pi$.

16 [Challenging] • Target: Area and average value • Cross-Lesson: 1.01 Functions of Several Variables & Partial Derivatives

Use a double integral to find the area of the region bounded by $y = x^2$ and $y = x$ (for $0 \le x \le 1$, where $x \ge x^2$).

Area $= \int_0^1\int_{x^2}^{x} 1\,dy\,dx = \int_0^1 (x - x^2)\,dx = \left[\frac{x^2}{2} - \frac{x^3}{3}\right]_0^1 = \frac{1}{2} - \frac{1}{3} = \frac{1}{6}$.

17 [Challenging] • Target: Area and average value • Cross-Lesson: 2.01 The Chain Rule & Directional Derivatives

The average value of $f(x, y)$ over a region D with area A is $\frac{1}{A}\iint_D f\,dA$. Find the average value of $f(x, y) = xy$ over the rectangle $[0, 2] \times [0, 3]$, using the earlier result $\iint_D xy\,dA = 9$.

Area of $[0, 2] \times [0, 3]$ is 6. Average value $= \frac{9}{6} = \frac{3}{2}$.

18 [Challenging] • Target: Area and average value • Cross-Lesson: 3.01 Optimization: Critical Points & Lagrange Multipliers

Use a double integral to find the area of the region bounded above by $y = 4 - x^2$ and below by $y = 0$.

The parabola meets $y = 0$ at $x = \pm 2$. Area $= \int_{-2}^{2}\int_0^{4-x^2} 1\,dy\,dx = \int_{-2}^{2}(4 - x^2)\,dx = \left[4x - \frac{x^3}{3}\right]_{-2}^{2} = \left(8 - \frac{8}{3}\right) - \left(-8 + \frac{8}{3}\right) = 16 - \frac{16}{3} = \frac{32}{3}$.

19 [Challenging] • Target: Volume via double integrals • Cross-Lesson: 1.01 Functions of Several Variables & Partial Derivatives

Set up (do not evaluate) the double integral for the volume under the surface $z = x^2 + y^2$ above the rectangle $[0, 1] \times [0, 1]$.

$V = \int_0^1\int_0^1 (x^2 + y^2)\,dy\,dx$.

20 [Challenging] • Target: Volume via double integrals • Cross-Lesson: 2.01 The Chain Rule & Directional Derivatives

Evaluate the volume integral set up above.

Inner: $\int_0^1 (x^2 + y^2)\,dy = x^2 y + \frac{y^3}{3}\big|_0^1 = x^2 + \frac{1}{3}$. Outer: $\int_0^1 \left(x^2 + \frac{1}{3}\right)dx = \frac{x^3}{3} + \frac{x}{3}\big|_0^1 = \frac{1}{3} + \frac{1}{3} = \frac{2}{3}$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Fubini's Theorem over Rectangles

Evaluate $\int_0^2 \int_0^3 2xy\,dy\,dx$.

a. 18 ✓

b. 9

c. 36

d. 12

Inner integral: $\int_0^3 2xy\,dy = [xy^2]_0^3 = 9x$. Outer integral: $\int_0^2 9x\,dx = [\frac{9}{2}x^2]_0^2 = \frac{9}{2}(4) = 18$.

22 [Foundational] • Target: Fubini's Theorem over Rectangles

What does Fubini's Theorem state for a continuous function $f(x, y)$ over a rectangle $R = [a, b] \times [c, d]$?

a. **The iterated integrals in either order are equal:** $\int_a^b \int_c^d f(x, y)\,dy\,dx = \int_c^d \int_a^b f(x, y)\,dx\,dy$ ✓

b. The double integral is always equal to $f(b, d) - f(a, c)$

c. The integral is zero if f is odd in x

d. The integral must be evaluated with x first

Fubini's Theorem guarantees that for continuous functions over rectangles, the order of integration does not affect the value of the double integral.

23 [Foundational] • Target: Fubini's Theorem over Rectangles

Evaluate $\iint_R (x + y)\,dA$ over $R = [0, 1] \times [0, 2]$.

a. 3 ✓

b. 2

c. 4

d. 1

$\int_0^1 \int_0^2 (x + y)\,dy\,dx = \int_0^1 [xy + \frac{y^2}{2}]_0^2\,dx = \int_0^1 (2x + 2)\,dx = [x^2 + 2x]_0^1 = 1 + 2 = 3$.

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24 [Foundational] • Target: Double Integrals over General Regions

Set up the double integral of $f(x, y)$ over the triangular region bounded by $y = 0$, $y = 2x$, and $x = 3$ as a Type I region.

a. $\int_0^3 \int_0^{2x} f(x, y)dydx$ ✓

b. $\int_0^6 \int_0^3 f(x, y)dxdy$

c. $\int_0^{2x} \int_0^3 f(x, y)dxdy$

d. $\int_0^3 \int_{2x}^6 f(x, y)dydx$

For a Type I region, x varies between constants $0 \le x \le 3$, and for each x, y goes from bottom boundary $y = 0$ to top boundary $y = 2x$: $\int_0^3 \int_0^{2x} f(x, y)dydx$.

25 [Foundational] • Target: Double Integrals over General Regions

Evaluate $\int_0^1 \int_0^x 6xydydx$.

a. $\frac{3}{4}$ ✓

b. 1

c. $\frac{1}{2}$

d. $\frac{3}{2}$

Inner: $\int_0^x 6xydy = [3xy^2]_0^x = 3x^3$. Outer: $\int_0^1 3x^3dx = [\frac{3}{4}x^4]_0^1 = \frac{3}{4}$.

26 [Foundational] • Target: Double Integrals over General Regions

What geometric quantity does $\iint_D 1dA$ compute?

a. **The area of the planar region** D ✓

b. The perimeter of region D

c. The volume under the paraboloid

d. The centroid

Integrating the constant height 1 over a 2D domain D calculates the area of D: $\text{Area}(D) = \iint_D dA$.

27 [Foundational] • Target: Changing the Order of Integration

To evaluate $\int_0^1 \int_y^1 e^{x^2}dxdy$, which cannot be integrated directly, reverse the order of integration.

a. $\int_0^1 \int_0^x e^{x^2}dydx$ ✓

b. $\int_0^1 \int_x^1 e^{x^2}dydx$

c. $\int_0^1 \int_0^1 e^{x^2}dxdy$

d. $\int_y^1 \int_0^1 e^{x^2}dydx$

Region: $0 \le y \le 1$ and $y \le x \le 1$. Inverting: $0 \le x \le 1$ and $0 \le y \le x$, yielding $\int_0^1 \int_0^x e^{x^2}dydx$.

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28 [Foundational] • Target: Changing the Order of Integration

Evaluate the reversed integral $\int_0^1 \int_0^x e^{x^2} dy dx$.

a. $\frac{e-1}{2}$ ✓

b. $e-1$

c. $\frac{e}{2}$

d. 1

Inner: $\int_0^x e^{x^2} dy = x e^{x^2}$. Outer: $\int_0^1 x e^{x^2} dx = [\frac{1}{2} e^{x^2}]_0^1 = \frac{1}{2}(e^1 - e^0) = \frac{e-1}{2}$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Changing the Order of Integration

Reverse the order of integration for $\int_0^4 \int_{\sqrt{x}}^2 f(x,y) dy dx$.

a. $\int_0^2 \int_0^{y^2} f(x,y) dx dy$ ✓

b. $\int_0^2 \int_{y^2}^4 f(x,y) dx dy$

c. $\int_0^4 \int_0^{y^2} f(x,y) dx dy$

d. $\int_{\sqrt{x}}^2 \int_0^4 f(x,y) dx dy$

Region: $0 \le x \le 4, \sqrt{x} \le y \le 2$. The curve is $y = \sqrt{x} \Rightarrow x = y^2$. For $y \in [0,2]$, x ranges from 0 to y^2: $\int_0^2 \int_0^{y^2} f(x,y) dx dy$.

30 [Intermediate] • Target: Double Integrals in Polar Coordinates

What is the differential area element dA in polar coordinates?

a. $dA = r dr d\theta$ ✓

b. $dA = dr d\theta$

c. $dA = r^2 dr d\theta$

d. $dA = \frac{1}{r} dr d\theta$

The Jacobian of the transformation $x = r\cos\theta, y = r\sin\theta$ is r, giving area element $dA = r dr d\theta$.

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31 [Intermediate] • Target: Double Integrals in Polar Coordinates

Evaluate $\iint_D e^{-(x^2+y^2)}dA$ where D is the disk $x^2+y^2 \le 4$.

a. $\pi(1-e^{-4})$ ✓

b. $2\pi(1-e^{-4})$

c. πe^{-4}

d. $\frac{\pi}{2}(1-e^{-4})$

In polar: $\int_0^{2\pi}\int_0^2 e^{-r^2} r\,dr\,d\theta = 2\pi[-\frac{1}{2}e^{-r^2}]_0^2 = 2\pi(-\frac{1}{2}e^{-4}+\frac{1}{2}) = \pi(1-e^{-4})$.

32 [Intermedi… …s in Polar Coordinates

Convert $\int_0^1\int_0^{\sqrt{1-x^2}} (x^2+y^2)dydx$ to polar coordinates.

a. $\int_0^{\pi/2}\int_0^1 r^3 dr d\theta$ ✓

b. $\int_0^{2\pi}\int_0^1 r^2 dr d\theta$

c. $\int_0^{\pi/2}\int_0^1 r^2 dr d\theta$

d. $\int_0^1\int_0^1 r^3 dr d\theta$

The region is the quarter disk in Quadrant I ($0 \le \theta \le \pi/2$, $0 \le r \le 1$). The integrand is $x^2+y^2 = r^2$, and $dA = r dr d\theta$, giving $\int_0^{\pi/2}\int_0^1 r^2 \cdot r dr d\theta = \int_0^{\pi/2}\int_0^1 r^3 dr d\theta$.

33 [Intermediate] • Target: Double Integrals in Polar Coordinates

Evaluate the double integral of the function $f(x,y) = 2xy$ over the rectangular region $R = [0,2] \times [0,3]$ using Fubini's Theorem. Set up the iterated integral by integrating with respect to y first, then x.

a. **12** ✓

b. 18

c. 6

d. 24

To evaluate the double integral $\iint_R 2xy\,dA$ where $R = [0,2] \times [0,3]$, we can use Fubini's Theorem. We will set up the iterated integral by integrating with respect to y first, then x: $\int_0^2\int_0^3 2xy\,dy\,dx$ First, integrate with respect to y, treating x as a constant: $\int_0^3 2xy\,dy = [xy^2]_{y=0}^{y=3} = x(3^2) - x(0^2) = 9x$ Now, integrate the result with respect to x.

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34 [Intermediate] • Target: Double integrals over a rectangle

Consider the region R in the first quadrant bounded by the x-axis, the y-axis, and the line $y = -2x + 6$. We are interested in evaluating the double integral $\iint_R f(x, y)\,dA$ where $f(x, y) = x + y$. However, the integral is set up in polar coordinates as $\int_{\alpha}^{\beta}\int_{r_1(\theta)}^{r_2(\theta)} f(r\cos\theta, r\sin\theta)\cdot r\,dr\,d\theta$. If this integral is evaluated from $\theta = 0$ to $\theta = \frac{\pi}{2}$, and the inner integral bounds are $r_1(\theta) = 0$ and $r_2(\theta) = \frac{6}{\sin\theta + 2\cos\theta}$, and the outer integral is $\int_0^{\pi/2}\int_0^{\frac{6}{\sin\theta+2\cos\theta}} (r\cos\theta + r\sin\theta)\cdot r\,dr\,d\theta$, determine the value of α and β that correspond to the description of region R in polar coordinates for $0 \le \theta \le \frac{\pi}{2}$.

a. $\alpha = 0, \beta = \frac{\pi}{2}$ ✓

b. $\alpha = 0, \beta = \frac{\pi}{4}$

c. $\alpha = 0, \beta = \frac{3\pi}{4}$

d. $\alpha = \frac{\pi}{4}, \beta = \frac{\pi}{2}$

The region R is in the first quadrant, bounded by the x-axis ($y = 0$), the y-axis ($x = 0$), and the line $y = -2x + 6$. When $x = 0$, $y = 6$. When $y = 0$, $2x = 6$, so $x = 3$. The vertices of the region are $(0, 0), (3, 0), (0, 6)$. In polar coordinates, $x = r\cos\theta$ and $y = r\sin\theta$. The line $y = -2x + 6$ becomes $r\sin\theta = -2r\cos\theta + 6$. Rearranging for r, we get $r(\sin\theta + 2\cos\theta) = 6$, so $r = \frac{6}{\sin\theta + 2\cos\theta}$. The first quadrant is defined by.

35 [Intermediate] • Target: Iterated integrals

Consider the region R in the xy-plane defined by $0 \le x \le 1$ and $0 \le y \le x^2$. Evaluate the double integral $\iint_R y\,dA$.

a. $\frac{1}{12}$ ✓

b. $\frac{1}{10}$

c. $\frac{1}{8}$

d. $\frac{1}{6}$

The double integral can be evaluated using Fubini's Theorem as an iterated integral. The region of integration is described by $0 \le x \le 1$ and $0 \le y \le x^2$. Therefore, the integral is set up as: $\int_0^1\int_0^{x^2} y\,dy\,dx$ First, we integrate with respect to y: $\int_0^{x^2} y\,dy = \left[\frac{y^2}{2}\right]_0^{x^2} = \frac{(x^2)^2}{2} - \frac{0^2}{2} = \frac{x^4}{2}$ Now, we integrate the result with respect to x.

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36 **[Intermediate]** • **Target: Fubini's Theorem: order of integration**

Consider the solid region E bounded below by the cone $z = \sqrt{x^2 + y^2}$ and above by the sphere $x^2 + y^2 + z^2 = 16$. The volume of E is given by the triple integral $V = \iiint_E dV$. If E is expressed in spherical coordinates, the integral becomes $\int_{\alpha}^{\beta} \int_{\gamma}^{\delta} \int_{\rho_1}^{\rho_2} \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta$. Determine the value of α, the lower limit of integration for θ.

a. **0** ✓

b. $\frac{\pi}{2}$

c. π

d. 2π

The region E is a spherical cone. The upper bound is a sphere of radius 4 centered at the origin, so ρ ranges from 0 to 4. The lower bound is the cone $z = \sqrt{x^2 + y^2}$. In spherical coordinates, $z = \rho\cos\phi$ and $\sqrt{x^2 + y^2} = \rho\sin\phi$. Thus, the cone equation becomes $\rho\cos\phi = \rho\sin\phi$. For $\rho eq 0$, this simplifies to $\cos\phi = \sin\phi$, which means $\tan\phi = 1$. In the range $0 \le \phi \le \pi$, this occurs at $\phi = \frac{\pi}{4}$. Therefore.

37 **[Intermediate]** • **Target: Double integrals over general regions**

Consider the vector field $\mathbf{F}(x, y, z) = \langle x, y, z \rangle$. Let S be the closed surface of the unit cube defined by $0 \le x \le 1, 0 \le y \le 1, 0 \le z \le 1$, oriented outwards. According to the Divergence Theorem, the flux of $\mathbf{F}$ through S is equal to the triple integral of the divergence of $\mathbf{F}$ over the volume V enclosed by S. Calculate this flux by evaluating $\iiint_V abla \cdot \mathbf{F} \, dV$.

a. 1

b. 2

c. **3** ✓

d. 4

The vector field is $\mathbf{F}(x, y, z) = \langle x, y, z \rangle$. The divergence of $\mathbf{F}$ is $abla \cdot \mathbf{F} = \frac{\partial}{\partial x}(x) + \frac{\partial}{\partial y}(y) + \frac{\partial}{\partial z}(z) = 1 + 1 + 1 = 3$. The volume V enclosed by the unit cube is given by $0 \le x \le 1, 0 \le y \le 1, 0 \le z \le 1$. The volume of this cube is $1 \times 1 \times 1 = 1$. Therefore, the flux is given by the triple integral: $\iiint_V abla \cdot \mathbf{F} \, dV = \iiint_V 3 \, dV$. Since.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Reversing the order of integration • Cross-Lesson: 3.01 Optimization: Critical Points & Lagrange Multipliers

Consider the solid region E in the first octant bounded by the coordinate planes and the surface $z = 4 - x^2 - y^2$. The volume of this region is given by the triple integral $V = \iiint_E dV$. We are given that the volume can also be expressed in cylindrical coordinates as $V = \int_0^{\pi/2} \int_0^{\sqrt{a}} \int_0^{4-r^2} r\,dz\,dr\,d\theta$. Find the value of the parameter a.

a. 2

b. **4** ✓

c. 16

d. 8

The solid region E is in the first octant and is bounded by $z = 0$, $x = 0$, $y = 0$ and $z = 4 - x^2 - y^2$. In cylindrical coordinates, $x^2 + y^2 = r^2$, so the surface is $z = 4 - r^2$. Since the region is in the first octant, $0 \le \theta \le \pi/2$. The lower bound for z is $z = 0$. The upper bound for z is $4 - r^2$. \\ To determine the bounds for r, we look at the projection of the solid onto the xy-plane. This projection is determined by the intersection of $z = 4 - x^2 - y^2$ with the xy-p.

39 [Challenging] • Target: Polar coordinates • Cross-Lesson: 1.01 Functions of Several Variables & Partial Derivatives

Consider the solid region E defined by $0 \le x \le 1$, $0 \le y \le 2$, and $0 \le z \le 3$. Evaluate the triple integral of the function $f(x, y, z) = 2x + 3y$ over the region E.

a. **18** ✓

b. 15

c. 21

d. 24

The triple integral of $f(x, y, z) = 2x + 3y$ over the rectangular box E is given by: $\iiint_E (2x + 3y)\,dV = \int_0^3 \int_0^2 \int_0^1 (2x + 3y)\,dx\,dy\,dz$ First, integrate with respect to x: $\int_0^1 (2x + 3y)\,dx = [x^2 + 3yx]_0^1 = (1^2 + 3y(1)) - (0^2 + 3y(0)) = 1 + 3y$ Next, integrate with respect to y: $\int_0^2 (1 + 3y)\,dy = \left[y + \frac{3}{2}y^2\right]_0^2 = (2 + \frac{3}{2}(2)^2) - (0 + \frac{3}{2}(0)^2) = 2 + \frac{3}{2}(4) = 2 + 6 = 8$.

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40 [Challenging] • Target: Area and average value • Cross-Lesson: 2.01 The Chain Rule & Directional Derivatives

A solid region E is defined by the inequalities $x^2 + y^2 \leq 16$, $0 \leq z \leq 5$. The volume of this region can be computed using a triple integral. In cylindrical coordinates, the region E is described by $0 \leq r \leq R$, $0 \leq \theta \leq 2\pi$, and $0 \leq z \leq H$. What is the value of R?

a. 2

b. **4** ✓

c. 8

d. 16

The given region is a cylinder with radius R_{cyl} and height H_{cyl}. The inequality $x^2 + y^2 \leq 16$ describes a circle of radius 4 centered at the origin in the xy-plane. When converting to cylindrical coordinates, $x^2 + y^2$ becomes r^2. Thus, the inequality $r^2 \leq 16$ implies $r \leq 4$. Therefore, in cylindrical coordinates, the radial component of the region is described by $0 \leq r \leq 4$. The question states that the region is described by $0 \leq r \leq R$, so $R = 4$. The inequality.

41 [Challenging] • Target: Volume via double integrals • Cross-Lesson: 3.01 Optimization: Critical Points & Lagrange Multipliers

Consider the region in the first octant bounded by the coordinate planes and the plane $x + y + z = 4$. If the density of the region is given by $\rho(x, y, z) = 3xy$, set up the triple integral for the mass of this region in Cartesian coordinates.

a. $\int_0^4 \int_0^{4-x} \int_0^{4-x-y} 3xy \, dz \, dy \, dx$ ✓

b. $\int_0^4 \int_0^{4-y} \int_0^{4-y-z} 3xy \, dx \, dz \, dy$

c. $\int_0^4 \int_0^{4-x} \int_0^{4-x-y} 3 \, dz \, dy \, dx$

d. $\int_0^4 \int_0^{4-z} \int_0^{4-z-y} 3xy \, dx \, dy \, dz$

The region is in the first octant, bounded by $x = 0$, $y = 0$, $z = 0$ and $x + y + z = 4$. We can express the bounds for z in terms of x and y from the plane equation: $z = 4 - x - y$. Since it's in the first octant, z goes from 0 to $4 - x - y$. The projection of this region onto the xy-plane is a triangle bounded by $x = 0$, $y = 0$ and $x + y = 4$. Thus, y ranges from 0 to $4 - x$, and x ranges from 0 to 4. The density is $\rho(x, y, z) = 3xy$. Therefore, the triple integral for the mass is.

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42 [Challenging] • Target: Double integrals in applications: mass and center of mass • Cross-Lesson: 1.01 Functions of Several Variables & Partial Derivatives

Consider the vector field $\mathbf{F}(x, y) = \langle x^2, y \rangle$. According to Green's Theorem, the line integral of $\mathbf{F}$ around a simple closed curve C is equal to the double integral of $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$ over the region D enclosed by C, where $\mathbf{F} = \langle P, Q \rangle$. For the given vector field $\mathbf{F}(x, y) = \langle x^2, y \rangle$, identify the value of $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$.

a. **0** ✓

b. x^2

c. y

d. -2x

The given vector field is $\mathbf{F}(x, y) = \langle P(x, y), Q(x, y) \rangle = \langle x^2, y \rangle$. According to Green's Theorem, we need to compute $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$. First, find the partial derivative of Q with respect to x: $\frac{\partial Q}{\partial x} = \frac{\partial}{\partial x}(y) = 0$. Next, find the partial derivative of P with respect to y: $\frac{\partial P}{\partial y} = \frac{\partial}{\partial y}(x^2) = 0$. Therefore.

43 [Challenging] • Target: Fubini's Theorem over Rectangles • Cross-Lesson: 2.01 The Chain Rule & Directional Derivatives

Consider the surface S which is the portion of the plane $z = 2x + 3y + 1$ lying above the rectangular region $R = [0, 1] \times [0, 2]$ in the xy-plane. Calculate the surface area of S.

a. $\sqrt{14}$

b. $2\sqrt{14}$ ✓

c. $3\sqrt{14}$

d. $4\sqrt{14}$

The surface area of a surface $z = g(x, y)$ above a region R is given by the integral $\iint_R \sqrt{1 + (g_x)^2 + (g_y)^2}\, dA$. Here, $g(x, y) = 2x + 3y + 1$. The partial derivatives are $g_x = 2$ and $g_y = 3$. Therefore, $\sqrt{1 + (g_x)^2 + (g_y)^2} = \sqrt{1 + 2^2 + 3^2} = \sqrt{1 + 4 + 9} = \sqrt{14}$. The integral becomes $\iint_R \sqrt{14}\, dA$, where $R = [0, 1] \times [0, 2]$. This evaluates to $\sqrt{14} \times \text{Area}(R)$. The area of the rectangle R is $(1 - 0) \times (2 - 0) = 2$. So, the sur.

44 [Challenging] • Target: Double Integrals over General Regions • Cross-Lesson: 3.01 Optimization: Critical Points & Lagrange Multipliers

Use spherical coordinates to find the volume of the upper half of the ball $x^2 + y^2 + z^2 \leq 16$.

a. $\frac{256\pi}{3}$

b. $\frac{64\pi}{3}$

c. $\frac{128\pi}{3}$ ✓

d. 32π

$dV = \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta$; integrating over the region gives $\frac{1}{2} \cdot \frac{4}{3}\pi(4)^3 = \frac{128\pi}{3}$.

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45 [Challenging] • Target: Changing the Order of Integration • Cross-Lesson: 1.01 Functions of Several Variables & Partial Derivatives

Use spherical coordinates to find the volume of the upper half of the ball $x^2 + y^2 + z^2 \le 4$.

a. $\frac{16\pi}{3}$ ✓

b. $\frac{32\pi}{3}$

c. $\frac{8\pi}{3}$

d. 4π

$dV = \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta$; integrating over the region gives $\frac{1}{2} \cdot \frac{4}{3}\pi(2)^3 = \frac{16\pi}{3}$.

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